

Platform Membership and Marketing-Mix Effectiveness

Formal Feature Definitions, Model Tests, Results, and Interpretation

JAMS Platform Membership Analysis

May 15, 2026

1 Research Objective

The empirical question is whether membership in a mobility platform changes how marketing-mix instruments translate into provider performance. The theoretical point is not that platforms should mechanically increase utilisation. A platform can create demand, but it can also intensify comparison, increase price salience, and expose weak elements of the provider’s offer. The analysis therefore treats platform membership as a market-interface condition that can moderate the effect of price, promotion, place, product, and process features.

The empirical setting is a city–provider–day panel from March 11, 2024 to August 22, 2024. The main analysis uses 14,514 observations from 97 city–provider units across 30 cities. The primary outcome is transformed capacity utilisation. For selected robustness tests, we also use transformed trip activity as a secondary outcome. Platform membership is defined as substantive bookable/payable integration in a platform interface; weaker forms of exposure such as pure routing information or visibility are treated separately.

2 Notation

Let i index providers, c cities, t calendar days, and $m(i)$ the provider’s mobility mode. The outcome is Y_{ict} , transformed capacity utilisation for provider i in city c on day t . The secondary outcome is Q_{ict} , transformed trip activity.

Platform membership is denoted by P_{ict} . We distinguish three mutually informative platform architectures:

$$P_{ict}^A = \text{large aggregator only}, \quad P_{ict}^L = \text{local MaaS only}, \quad P_{ict}^M = \text{multi-platform membership}.$$

The omitted comparison group consists of provider–city–days without clean bookable/payable platform membership.

Marketing-mix variables are grouped as follows:

$$\mathcal{M}_{ict} = \{U_{ict}, R_{ict}, H_{ict}, B_{ict}, V_{ict}, C_{ict}, E_{ict}, A_{ict}, N_{ict}\}.$$

Here, U_{ict} is the unlock fee, R_{ict} is the per-minute price, H_{ict} is promotion activity, B_{ict} is product breadth, V_{ict} is relative active-vehicle availability, C_{ict} is direct same-mode competition, E_{ict} is same-platform competition, A_{ict} captures commute-oriented usage, and N_{ict} captures night-oriented usage.

All continuous moderator variables used in interactions are standardised before estimation, except for binary threshold and promotion indicators. Standardisation makes interaction coefficients comparable across feature families.

3 Feature Definitions

3.1 Platform Architecture

Clean platform membership equals one when a provider can be substantially accessed through the platform interface. The coding requires bookable/payable integration or comparable substantive transaction integration. Weak visibility, route information, or unclear exposure is not treated as full membership. Platform architecture is then split into large aggregator membership, local public MaaS membership, and simultaneous membership in both types.

3.2 Pricing Features

The absolute unlock-fee threshold feature is

$$D_{ict}^{\tau} = \mathbb{1}\{U_{ict} \geq \tau\},$$

where τ is a salient threshold such as EUR 1.00 or EUR 1.20. This feature captures whether the provider charges an upfront access fee high enough to become salient before the ride begins.

The relative same-mode unlock-fee disadvantage is defined as

$$\Delta_{ict}^U = z\left(U_{ict} - \min_{j \in \mathcal{S}_{ct}(m(i))} U_{jct}\right),$$

where $\mathcal{S}_{ct}(m(i))$ is the set of providers in city c on day t that operate in the same mobility mode as provider i , and $z(\cdot)$ denotes standardisation. If a provider is the cheapest same-mode option, the raw gap is zero. This feature operationalises relative access-price disadvantage: the provider is not merely expensive in absolute terms, but expensive relative to directly comparable alternatives.

Analogously, the relative per-minute price disadvantage is

$$\Delta_{ict}^R = z\left(R_{ict} - \min_{j \in \mathcal{S}_{ct}(m(i))} R_{jct}\right).$$

This parallel feature allows us to test whether platform membership amplifies all price differences or specifically upfront price salience.

3.3 Competition and Availability

Direct same-mode competition is measured as

$$C_{ict} = |\mathcal{S}_{ct}(m(i))| - 1,$$

standardised for modelling. It counts how many directly comparable same-mode alternatives are active in the same city on the same day.

Same-mode active-vehicle share is

$$V_{ict} = \frac{Fleet_{ict}}{\sum_{j \in \mathcal{S}_{ct}(m(i))} Fleet_{jct}},$$

where $Fleet_{ict}$ is the active vehicle count for provider i in city c on day t . This captures the provider's relative availability position among direct substitutes.

Same-platform competition counts providers that are visible or bookable through the same platform environment. This feature is theoretically important because platforms can create within-interface comparison. Empirically, however, it is tightly linked to platform architecture and city/provider structure, so it is treated cautiously.

3.4 Product, Promotion, and Process Features

Product breadth, B_{ict} , is the standardised number of vehicle types offered by the provider in the city–day. The interpretation is straightforward: broader assortment can increase modal fit and may buffer pricing frictions.

Promotion activity, H_{ict} , captures whether a provider–city–day is covered by an identified promotion. We distinguish broad promotion activity from more specific usage-cost reducing promotions, such as free minutes for public-transport subscribers. These specific promotions are theoretically relevant because they reduce marginal usage cost or offset access-price salience, but their current empirical support is localized.

Commute-oriented usage and night-oriented usage are derived from the distribution of observed trips over the day. The night-share feature, N_{ict} , captures whether provider usage is disproportionately concentrated in evening and night contexts. This is a process/context feature: it indicates the kind of usage situation in which the service is being consumed.

Provider-level product/process affordances include reservation availability, pass/free-unlock availability, minute-bundle availability, and parking/geofence rules. These are useful for describing the marketing mix, but current variation is mostly provider-level rather than city–date specific. They are therefore treated as theory-supporting features unless future data provide historical city-level timing.

4 Model Specification

We estimate fixed-effect models with cluster-robust standard errors at the city–provider level. Two fixed-effect structures are used throughout:

$$\alpha_{ict}^M = \lambda_c + \mu_i + \delta_t$$

for market fixed effects, and

$$\alpha_{ict}^U = \psi_{ic} + \delta_t$$

for unit fixed effects. The market specification controls for city, provider, and day differences. The unit specification is stricter because it identifies within city–provider variation over time, net of day shocks.

4.1 Platform Main Effect

The baseline platform model is

$$Y_{ict} = \beta P_{ict} + X'_{ict}\Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

This model asks whether platform membership is associated with a uniform utilisation shift. It is intentionally not the central theory test; it is a diagnostic benchmark.

4.2 Marketing-Mix Moderation

The general moderation model is

$$Y_{ict} = \beta_1 M_{ict} + \beta_2 P_{ict} \beta_3 (M_{ict} \times P_{ict}) + X'_{ict}\Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

The coefficient β_3 tests whether platform membership changes the effectiveness of a marketing-mix feature M_{ict} .

4.3 Platform-Architecture Moderation

For platform types, we estimate

$$Y_{ict} = \sum_{k \in \{A, L, M\}} \theta_k (M_{ict} \times P_{ict}^k) + X'_{ict} \Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

The key terms are θ_L and θ_M , because the theory predicts stronger comparison effects in local MaaS and multi-platform contexts than in large aggregator-only contexts.

4.4 High Unlock-Fee Threshold Models

The threshold model is

$$Y_{ict} = \sum_{k \in \{A, L, M\}} \gamma_{k\tau} (D_{ict}^\tau \times P_{ict}^k) + X'_{ict} \Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

Here $\gamma_{k\tau}$ tests whether a salient upfront fee threshold is especially harmful inside platform architecture k .

4.5 Relative Price-Position Models

The relative price-position model is

$$Y_{ict} = \sum_{k \in \{A, L, M\}} \eta_k (\Delta_{ict}^U \times P_{ict}^k) + X'_{ict} \Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

This is the cleanest empirical test of the comparison-salience mechanism: if platforms make direct alternatives more comparable, then being more expensive than same-mode competitors should be particularly harmful in platform contexts.

4.6 Product-Buffer Model

To test whether product breadth buffers pricing friction, we estimate

$$Y_{ict} = \sum_{k \in \{A, L, M\}} \rho_k (D_{ict}^{1.00} \times P_{ict}^k \times B_{ict}) + X'_{ict} \Gamma + \alpha_{ict} + \varepsilon_{ict}.$$

A positive ρ_L means that product breadth weakens the Local MaaS high-fee penalty.

5 Results

Table 1: Relevant Empirical Findings

Finding	Substantive result	Robustness	Market fixed effects	Unit fixed effects	Empirical basis
F1	Platform membership does not create a stable generic utilisation lift.	Not robust as a main effect	0.061; $p = 0.008$	-0.010; $p = 0.205$	Full sample, $N = 14,514$. The positive association disappears in the stricter within-unit model.
F2	Unlock fees become more detrimental in platform contexts.	Robust in market models; weaker within-unit	-0.0715; $p = < 0.001$	-0.0236; $p = 0.349$	Full sample, $N = 14,514$. Repeated negative evidence across concept and pricing models.
F3	High unlock fees are harmful in platform contexts.	Robust	-0.104; $p = < 0.001$	-0.040; $p = < 0.001$	Full sample, $N = 14,514$. Negative in all six threshold specifications.
F3/F4	High unlock fees are especially harmful in Local MaaS contexts.	Very robust	-0.150; $p = 0.002$	-0.046; $p = < 0.001$	Full sample, $N = 14,514$. Stable across threshold and platform-type model families.
F3/F4	High unlock fees are especially harmful in multi-platform contexts.	Very robust	-0.131; $p = 0.007$	-0.098; $p = < 0.001$	Full sample, $N = 14,514$. Stable across threshold and platform-type model families.
F3	EUR 1.00 threshold in Local MaaS contexts.	Very robust	-0.229; $p = < 0.001$	-0.049; $p = 0.002$	Full sample, $N = 14,514$; 90 Local MaaS high-fee observations.
F3	EUR 1.00 threshold in multi-platform contexts.	Very robust	-0.146; $p = < 0.001$	-0.099; $p = < 0.001$	Full sample, $N = 14,514$; 1,308 multi-platform high-fee observations.
F3	EUR 1.20 threshold in Local MaaS contexts.	Very robust	-0.233; $p = < 0.001$	-0.047; $p = 0.006$	Full sample, $N = 14,514$; 72 Local MaaS high-fee observations.
F3	EUR 1.20 threshold in multi-platform contexts.	Very robust	-0.171; $p = < 0.001$	-0.082; $p = < 0.001$	Full sample, $N = 14,514$; 1,029 multi-platform high-fee observations.
F3a	The Local MaaS high-fee penalty is not driven by one city.	Robust	Median city-drop estimate at EUR 1.00: -0.209	Median city-drop estimate at EUR 1.00: -0.035	All 13 leave-one-platform-city tests remain negative and significant for EUR 0.99, EUR 1.00, and EUR 1.20.
F3a	The multi-platform high-fee penalty is not a single-city artifact.	Robust at EUR 1.00 and EUR 1.20	Median city-drop estimates: -0.107 at EUR 1.00 and -0.135 at EUR 1.20	Median city-drop estimates: -0.085 at EUR 1.00 and -0.073 at EUR 1.20	All 13 leave-one-platform-city tests remain negative and significant at the EUR 1.00 and EUR 1.20 thresholds.
F4a	Large aggregator-only membership does not show a stable high-fee penalty.	Negative evidence	-0.045; $p = 0.066$	-0.016; $p = 0.325$	Full sample, $N = 14,514$. The high-fee effect is not a large-aggregator-only result.
F3b	The Local MaaS high-fee result generalises to trip activity.	Robust for Local MaaS	-1.014; $p = < 0.001$	-0.130; $p = < 0.001$	Full sample, $N = 14,514$. The result is not merely a capacity-ratio artifact.
F3b	Multi-platform evidence is cleaner for utilisation than for trip activity.	Moderate	-0.538; $p = < 0.001$	-0.010; $p = 0.873$	Full sample, $N = 14,514$. Strong in market models, not within unit.

Finding	Substantive result	Robustness	Market fixed effects	Unit fixed effects	Empirical basis
F5	Promotion in platform contexts is positive mainly within provider–city units.	Moderate	0.012; $p = 0.424$	0.0115; $p = < 0.001$	Full sample, $N = 14,514$; 738 promotion observations.
F6	Subscriber-free-minutes promotions fit multi-platform exposure.	Exploratory mechanism evidence	0.122; $p = < 0.001$	0.043; $p = < 0.001$	Full sample model, $N = 14,514$, but treated support is only one city–provider cell with 144 observations.
F16	Vehicle-type breadth buffers the Local MaaS high-fee penalty.	Moderate to robust	0.074; $p = < 0.001$	0.014; $p = < 0.001$	Full sample, $N = 14,514$; 90 Local MaaS high-fee observations.
F19	Relative same-mode unlock-fee disadvantage is harmful in Local MaaS contexts.	Very robust; strongest new feature result	-0.089; $p = < 0.001$	-0.030; $p = 0.002$	Full sample, $N = 14,514$. The feature is available for the full panel.
F19	Relative same-mode unlock-fee disadvantage is harmful in multi-platform contexts.	Very robust	-0.091; $p = < 0.001$	-0.041; $p = < 0.001$	Full sample, $N = 14,514$. The feature is available for the full panel.
F20	Relative per-minute price disadvantage is weaker than relative unlock-fee disadvantage.	Moderate	-0.104; $p = 0.003$	-0.020; $p = 0.153$	Full sample, $N = 14,514$. Supports upfront-fee salience rather than generic price sensitivity.
F21	Same-mode competition is a Local MaaS boundary condition.	Moderate	-0.059; $p = 0.013$	-0.019; $p = 0.083$	Full sample, $N = 14,514$. Weaker than the relative unlock-fee gap.
F22	Own same-mode availability helps in some platform contexts but does not offset high fees.	Moderate	-0.007; $p = 0.819$ for multi-platform	0.019; $p = 0.024$ for multi-platform	Full sample, $N = 14,514$. Positive mainly within unit; high-fee triples remain problematic.
F23	Local MaaS is less aligned with night-oriented usage.	Moderate	-0.019; $p = 0.001$	-0.018; $p = < 0.001$	Full sample, $N = 14,514$. Usage-context evidence from observed trip timing.
F24	Provider-level pass, free-unlock, reservation, and parking/geofence features are not robust estimators yet.	Identification caveat	Not interpreted because of collinearity	Not interpreted because of weak within-unit variation	The features are useful for theory and future coding, but current support is mostly provider-level.
F25	Same-platform competitor count is not estimable cleanly.	Not supported	Implausibly large coefficients and missing p-values	Unstable and collinear	The measure is too mechanically tied to platform architecture and city/provider structure.

6 Description of the Evidence

The first empirical pattern is that platform membership is not a reliable main-effect story. The market fixed-effect estimate is positive, but the effect disappears in the stricter provider–city fixed-effect specification. This motivates the moderation logic: platforms matter less as an independent treatment that raises utilisation everywhere and more as an interface that changes how users evaluate the provider’s marketing mix.

The second and most stable pattern is pricing friction. High upfront access fees are consistently more harmful in platform contexts, especially in Local MaaS and multi-platform settings. At the EUR 1.00 threshold, Local MaaS has a market fixed-effect coefficient of -0.229 and a unit fixed-effect coefficient of -0.049. Multi-platform membership has a market fixed-effect coefficient of -0.146 and a unit fixed-effect coefficient of -0.099. The EUR 1.20 threshold produces the same

qualitative conclusion.

The robustness checks strengthen this interpretation. The Local MaaS high-fee penalty survives every leave-one-platform-city check at EUR 0.99, EUR 1.00, and EUR 1.20. The multi-platform high-fee penalty survives every city-drop test at EUR 1.00 and EUR 1.20. The Local MaaS result also appears in trip activity, which indicates that the finding is not merely a mathematical artefact of the utilisation denominator.

The third pattern is that relative price position clarifies the mechanism. Providers with higher unlock fees than their same-mode competitors perform worse in Local MaaS and multi-platform contexts. The relative unlock-fee disadvantage is negative and significant in both market and unit specifications. The analogous per-minute price disadvantage is weaker. This distinction is central: platforms do not simply make all price differences more important; they appear to make upfront access-price differences especially salient.

The fourth pattern is that other marketing-mix elements matter, but they are secondary. Product breadth buffers the Local MaaS high-fee penalty, suggesting that a richer product assortment can partially offset pricing friction by improving modal fit. Promotion shows positive within-unit evidence, and subscriber-free-minutes promotions fit the multi-platform logic, but the campaign evidence is localized. Same-mode competition is a useful boundary condition, yet it is weaker than relative price position. Finally, provider-level pass, reservation, and parking/geofence features are useful descriptive and theoretical variables, but current variation is not sufficient for robust estimation.

7 Interpretation and Narrative

The results support a comparison-salience account of platform membership. Platforms change the shopping and decision environment. They bring alternatives into the same interface, make providers easier to compare, and increase the visibility of differences that might otherwise be less salient. The central marketing consequence is that the same pricing decision can have different effects inside and outside platform environments.

The strongest evidence concerns upfront access fees. Unlock fees are discrete, immediate, and visible before the trip begins. They are therefore well suited to become salient in a platform interface. Per-minute prices require users to anticipate trip duration and total cost, which makes them less transparent at the point of choice. This explains why the relative unlock-fee gap produces the strongest new feature result, while the relative per-minute price gap is weaker.

Local MaaS and multi-platform contexts are the empirical core of the story. In these contexts, users are not merely exposed to one additional booking channel. They are exposed to a broader urban mobility choice architecture in which shared mobility competes with other providers and, often, other transport modes. The platform therefore acts as a comparison environment. A provider with a high or relatively high unlock fee becomes less attractive precisely because the platform makes alternatives more visible.

The negative evidence for large aggregator-only membership is important. If the findings were only a Free Now-style aggregator effect, the interpretation would be narrower and less theoretically distinctive. Instead, the strongest evidence appears in local public MaaS and multi-platform settings. This supports a broader claim about platform architecture: the effect depends on how the platform structures choice, not merely on whether a provider is present on a commercial aggregator.

The product-breadth result makes the narrative more balanced. Platform membership can amplify pricing frictions, but providers are not powerless. Broader vehicle-type offerings appear to buffer the Local MaaS high-fee penalty. This suggests an interaction between price and product strategy: a salient access fee is harmful, but greater modal fit can soften the effect. In marketing terms, platform participation raises the bar for internal consistency across the mix. Price, product breadth, and availability need to work together.

The promotion findings point in a similar direction but remain empirically narrower. Usage-cost reducing promotions such as subscriber free minutes fit the platform logic because they counteract cost salience. However, the observed campaign support is too localized for a central causal claim. The result should be framed as mechanism-consistent evidence and as a priority for future city-date campaign coding.

Competition should be interpreted as an enabling condition rather than as the main mechanism. Same-mode competition matters because it supplies alternatives that can be compared. But the stronger result is not the competitor count itself; it is the provider’s relative access-price disadvantage. This means the paper should not argue simply that competition moderates platform effects. A more precise claim is that platforms make marketing-mix disadvantages more consequential when direct alternatives are available and visible.

8 Implications

The empirical narrative can be stated as follows:

Platform membership reshapes marketing-mix effectiveness by changing the comparison environment in which users evaluate mobility providers. In Local MaaS and multi-platform contexts, high and relatively high unlock fees reduce capacity utilisation. The effect is not a generic platform main effect, not a large aggregator-only result, and not driven by a single city. Product breadth can buffer the penalty, while promotion and pass features remain promising but require richer historical campaign coding.

For the paper, the most defensible main results are: high unlock-fee thresholds in Local MaaS and multi-platform contexts; relative same-mode unlock-fee disadvantage; leave-one-city robustness; and trip-activity robustness for Local MaaS. Product breadth should be used as a secondary moderation result. Promotion and provider-level pass/process features should support the mechanism and motivate future data collection rather than anchor the main identification claim.

9 Limitations

The analysis is observational and estimates conditional associations rather than randomized treatment effects. Some platform architectures are structurally tied to specific cities and providers, which is why unit fixed effects and leave-one-city checks are essential. Several promising marketing-mix features, especially passes, reservations, and parking/geofence rules, currently vary mostly at the provider level and therefore cannot be interpreted as clean city-date effects. Promotion findings are based on limited campaign support. Same- platform competitor counts are theoretically interesting but currently too collinear with platform architecture to estimate cleanly.